≠ Inequalities

AN INEQUALITY IS USED TO SHOW THAT ONE QUANTITY IS NOT EQUAL TO ANOTHER.

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Inequality symbols

An inequality symbol shows that the numbers on either side of it are different in size and how they are different. There are five main inequality symbols. One simply shows that two numbers are not equal, the others show in what way they are not equal.

$$x \neq y$$

◁ **Not equal to**
This sign shows that x is not equal to y; for example, $3 \neq 4$.

$$x > y \qquad x \geqslant y \qquad x < y \qquad x \leqslant y$$

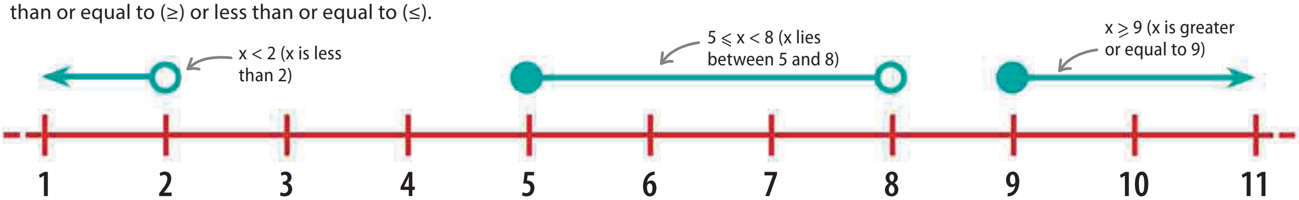
△ **Greater than**
This sign shows that x is greater than y; for example, $7 > 5$.

△ **Greater than or equal to**
This sign shows that x is greater than or equal to y.

△ **Less than**
This sign shows that x is less than y. For example, $-2 < 1$.

△ **Less than or equal to**
This sign shows that x is less than or equal to y.

▽ **Inequality number line**
Inequalities can be shown on a number line. The empty circles represent greater than (>) or less than (<), and the filled circles represent greater than or equal to (≥) or less than or equal to (≤).



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Rules for inequalities

Inequalities can be rearranged, as long as any changes are made to both sides of the inequality. If an inequality is multiplied or divided by a negative number, then its sign is reversed.

▷ **Multiplying or dividing by a positive number**
When an inequality is multiplied or divided by a positive number, its sign does not change.

$$a \geqslant 4 \xrightarrow{\times +3} 3a \geqslant 12$$
$$a \geqslant 4 \xrightarrow{\div +4} \frac{a}{4} \geqslant 1$$

sign stays the same

4 added to both sides of sign

$$x < -4 \xrightarrow{+4} x + 4 < 0$$
$$x < -4 \xrightarrow{-2} x - 2 < -6$$

sign stays the same

2 subtracted from both sides of sign

△ **Adding and subtracting**
When an inequality has a number added to or subtracted from it, its sign does not change.

$$p < 3 \xrightarrow{\times -3} -3p > -9$$
$$p < 3 \xrightarrow{\div -1} -p > -3$$

sign is reversed

△ **Multiplying or dividing by a negative number**
When an inequality is multiplied or divided by a negative number, its sign is reversed. In this example, a less than sign becomes a greater than sign.